

Sandipan Dey

Software Engineer – Rust, Bitcoin, Backend Systems, Infrastructure

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PROFILE

Software engineer with 6+ years building resilient systems for financial and Bitcoin applications. Focused on Rust backends, event-sourced architectures, reproducible infrastructure, and developer tooling.

EXPERIENCE

Galoy, Remote

Software Engineer

Jul 2022 – Present

- Contributed credit-facility updates for collateral event publishing, activation terminology, and active-loan reporting.
- Added lifecycle test coverage for credit facilities, including scenario fixtures and end-to-end validations.
- Updated accounting configuration for chart templates, FX/deposit setup, fiscal periods, and ledger/report edge cases.
- Supported reporting delivery through admin UI/API changes, signed downloads, webhook authentication, and fixes.
- Improved data/reporting jobs using Dagster, PostgreSQL, BigQuery, price snapshots, and validation checks.
- Contributed [Blink Wallet](#) mobile features across Circles, notifications, bulletins, theming, and payment screens.
- Supported mobile quality and release tooling with E2E tests, store delivery, reproducible builds, CI/CD, and Helm.

Intern

Oct 2021 – Jun 2022

- Supported Bitcoin and Lightning infrastructure while building Concourse CI improvements and Kubernetes deployment patterns for developer workflows.

Bitcoin Dev Kit, Remote – **Open Source Contributor, Summer of Bitcoin**

Jun 2021 – Sep 2021

- Added Bech32m wallet-send test coverage in bdk and fixed popular-address indexing in bitcoin-regtest-box.
- Improved BDK tooling with verbose PSBT/transaction inspection in bdk-cli and upgraded rust-bitcoin in rust-electrum-client.
- Wrote technical articles on Bitcoin internals: co-authored [The Hidden Power of Bitcoin](#) on Miniscript/descriptors and wrote [Curves Behind Bitcoin](#) on secp256k1, ECDSA, and Schnorr signatures.

Durbin Technologies & Quinch Systems, Kolkata, India – **Associate Software Engineer**

Oct 2019 – Oct 2021

- Built OAuth2-based in-house authentication across internal products.
- Delivered logistics/HR software for 25 vehicles/workers across 3 clients, with asset tracking, payroll, and billing.
- Created **xIngest**, an in-house IoT framework for millions of parallel device events, live dashboards, and OTA updates.
- Integrated Razorpay, Twilio, and Amazon SNS for payments, messaging, and alerts.
- Built RailSwitch, an [OHE lines](#) control app for Indian Railways. RailSwitch controlled heavy-duty OHE isolator relays with ESP8266, MQTT, and React Native; deployed at Howrah Station and used for over 3 years.
- Built ventilator R&D interfaces for live sensor data during COVID-19.
- Built computer-vision based fire detection pipelines with Python, OpenCV, TensorFlow over industrial camera feeds.

TECHNICAL EXPERTISE

Languages

Rust, TypeScript, JavaScript, Bash, Python, C/C++

Backend & Data

Node.js, PostgreSQL, Redis, OpenTelemetry, Meltano, Dagster, WebRTC, MQTT

Cloud

Nix, Terraform, Kubernetes, Docker, Helm, GCP, Concourse CI, GitHub Actions

Domains

Bitcoin, Lightning Network, event sourcing, domain-driven design, ledgers, IoT, embedded systems

EDUCATION

Academy of Technology – B.Tech. Computer Science & Engineering

2018 – 2022

CGPA: 9.2/10.0; highest SGPA: 10.0.